EMPLOYEE360 AI

An Evidence-Driven Autonomous AI Framework for Intelligent Employee Lifecycle Management

Final-Year Engineering Project Assigned by Zensar HR | Master Architectural Primer

Autonomous Loop: Understand -> Plan -> Gather Evidence -> Check Policy -> Act -> Verify -> Audit

1. Executive Summary & Core Paradigm

Employee360 AI is an evidence-driven, configurable autonomous AI framework engineered to manage three foundational pillars of the corporate employee lifecycle: (1) Hiring & Candidate Assessment, (2) Employee Onboarding, and (3) Employee Assistance & HR Case Resolution. Traditional HR chatbots are shallow FAQ responders that lack agency, evidence tracking, and policy verification. Employee360 AI replaces them with a single shared autonomous agent core configured by declarative, versioned schemas.

* **Hiring & Assessment:** Voice AI interview, 'Defend Your Resume' claim extraction, competency evidence mapping, real-time follow-up probes, and automated interview-integrity proctoring.
* **Employee Onboarding:** Interactive joining checklists, policy Q&A, document collection, proactive reminders, and mock action completion.
* **Employee Assistance:** Grounded policy RAG, autonomous execution of authorized mock HR transactions (leave, benefits, IT tickets), two-phase action verification, and escalation.
* **Proof-Carrying Decisions:** No black-box scores. Every recommendation is backed by a verified chain: Requirement -> Evidence Quote -> Source Document -> Rule -> Confidence -> Reviewer.

2. Zensar Scope, Problem Statements & Constraints

The project addresses two connected problem statements with strict enterprise guardrails:

* **PS1: Conversational HR Agent (Primary):** Develop one configurable agent conducting natural spoken dialogue in English (sub-1.5s latency), retrieving grounded answers via RAG, handling multi-turn context, and managing interruption and silence.
* **PS2: Interview Proctoring (Secondary):** Integrate parallel multi-modal anomaly detection (candidate absence, multiple faces, gaze/head pose deviation, secondary voices). Strictly non-punitive: anomalies are logged with snapshots as reviewer audit evidence, never as automated disqualification.
* **Project Constraints:** English-only; synthetic data and zero live integrations with Zensar internal ERP/Workday; no telephony (browser WebSockets only); local-first execution (no-Docker native setup).

3. Target Roles and Access Control (RBAC)

* **HR Administrator:** Configures jobs, competencies, evaluation rubrics, knowledge bases, and generates secure candidate interview invite links.
* **Candidate:** Opens secure link using one-time cryptographic HMAC token (no employee account required); completes camera/mic test, consent, and voice interview.
* **Employee:** Authenticates to Self-Service portal; queries policy documents; submits onboarding checklist items; requests mock HR actions.
* **HR Reviewer / Mentor:** Inspects proof-carrying evaluation reports, clickable transcript citations, proctoring anomaly evidence snapshots, and system audit logs.

4. Seven-Layer System Architecture

Employee360 AI is structured as a high-performance modular monolith running locally on standard development hardware:

* **Layer 1: Client Interfaces:** Next.js 14 + React + TypeScript web application with dedicated portals: HR Command Center, Candidate Interview Studio, Employee Self-Service, and Adversarial Proctoring Lab.
* **Layer 2: API Gateway & Auth:** FastAPI gateway with JWT Bearer authentication for employees/admins and HMAC signed tokens for candidates; bi-directional WebSocket hub.
* **Layer 3: Workflow Compiler:** Pydantic-based HR Agent Compiler that validates declarative JSON/YAML schemas, generates state graphs, and binds competency rubrics.
* **Layer 4: Autonomous Agent Core:** LangGraph / deterministic state orchestrator, turn management, Evidence Sufficiency Engine, Grounded RAG, and Two-Phase Action Engine.
* **Layer 5: Proctoring Engine (PS2):** OpenCV + MediaPipe Face Mesh (468 3D landmarks for gaze and head-pose), face count detector, and acoustic voice activity detector.
* **Layer 6: Model Abstractions:** Pluggable provider interfaces for STT (faster-whisper), TTS (Kokoro/Edge-TTS), LLM (Google Gemini / Ollama Llama 3 / OpenAI), and dense embeddings.
* **Layer 7: Persistence & Audit:** PostgreSQL with pgvector (or zero-config SQLite local vector fallback), local media snapshot storage, and immutable audit ledger.

5. Database Schema & API Specifications

Core relational entities and their responsibilities:

* **users & workflow_versions:** RBAC credentials and immutable compiled workflow schemas with SHA-256 integrity hashes.
* **jobs & competencies:** Job requisitions linked to competency matrices, evaluation criteria, and rubric weights.
* **interview_sessions & turns:** Tracks candidate sessions, cryptographic tokens, sequential dialogue turns, audio clip paths, and turn latencies.
* **evidence_records & proctoring:** Verified candidate claims mapped to competencies (confidence 0-1) and proctoring anomaly events with frame snapshots.
* **hr_cases, actions & onboarding:** Employee support tickets, authorized mock action receipts (two-phase verified), and onboarding task checklists.
* **FastAPI API Routes:** /api/v1/auth, /api/v1/workflows, /api/v1/jobs, /api/v1/interviews, /api/v1/evidence, /api/v1/proctoring, /api/v1/knowledge, /api/v1/employee, /api/v1/audit, and WebSocket /ws/v1/interview/{session_id}.

6. Empirical Research & Experimental Labs

As a final-year engineering capstone, Employee360 AI produces measurable, reproducible research data:

* **Conversational Metrics:** Time-to-First-Token (TTFT < 600ms), End-to-End Spoken Latency (< 1.5s), Task Completion Rate, and Information Extraction F1.
* **Grounding & RAG Metrics:** Citation grounding percentage, hallucination frequency, and escalation precision.
* **Evidence Sufficiency Metrics:** Competency coverage score and follow-up probe effectiveness rate.
* **Adversarial Proctoring Lab:** Synthetic testbed evaluating precision, recall, and false alarm rate against phone glances, whisper coaching, absence, and seat switching.
* **Counterfactual Fairness Lab:** Synthetic attribute permutation (demographic indicators) holding technical claims constant to audit score stability.

7. Phase-Wise Roadmap & Local Execution Guide

The project executes strictly in sequence without premature complexity:

* **Phase 0 (Current):** Architecture Blueprint, Documentation Suite, and Master Blueprint PDF.
* **Phase 1:** FastAPI + Next.js 14 foundation, RBAC Auth, HR Agent Compiler, and Seed Data.
* **Phase 2:** Spoken voice interview, WebSocket streaming, faster-whisper STT, TTS, and transcript persistence.
* **Phase 3:** Grounded RAG, 'Defend Your Resume', Evidence Sufficiency Engine, and Proof-Carrying Reports.
* **Phase 4:** Interview integrity proctoring (OpenCV/MediaPipe) and Adversarial Proctoring Lab.

* **Phase 5:** Autonomous Onboarding & Employee Assistance, Controlled Action Registry, and Two-Phase Verifier.
* **Phase 6 & 7:** Research evaluation benchmarks, Counterfactual Fairness Lab, testing, and final packaging.

Local Run Commands: Backend: 'uvicorn backend.main:app --reload --port 8000' | Frontend: 'npm run dev' (port 3000).